Ngoc Dung Nguyen

Curriculum Vitae

Vietnamese
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Tertiary Education

2018–2021 **Ph.D. student in Statistics**, *Department of Statistical Sciences, University of Padova*, Italy.
Advisor: Professor Alberto Roverato.
Thesis title: Structure learning of colored graphical models for paired data.

2017–2018 **Master in Science, Technology and Health**, *Université de Tours*, France, GPA: 16.796/20.
Master 2 Program of Mathematics. Major in Applied Mathematics. Mention: *"Très Bien"*.
Advisor: Professor Romain Abraham.
Thesis title: Limits or large random graphs.

2013–2017 **Bachelor in Science**, *Vietnam National University-Ho Chi Minh Univeristy of Science*, Ho Chi Minh City, Vietnam, GPA: 8.21/10.
Program in Mathematics and Computer Science. Major in Probability and Statistics. Mention: Very Good.
Advisor: Professor Dang Duc Trong.
Thesis title: Multiplicative censoring models.

Research Interests

Statistical learning Supervised and unsupervised learning; model inference, e.g. bootstrap methods, Bayesian methods; high-dimensional problems, e.g. regularization, principal component analysis (PCA), etc.

Applied Statistics Statistical inference, statistical modelling, regression analysis and diagnostics, model selection on model spaces, e.g. greedy searching.

Graphical Models Structure learning and model search for (undirected) graphical models, e.g. graphical Lasso, model selection of colored graphical models for paired data, inferring the distributions of quantities of interests given the observations.

Biostatistics Statistical methods and models for large and complex biological data sets, analysis of gene expression data, inference for biological/genetic networks.

Publications

Model selection in graphical models

2021 Working paper: Model selection for colored graphical models for paired data.

Conference, Seminar, and Workshop Presentations

Jul. 2019 Data Research Camp of the Department of Statistical Sciences of the University of Padova supported by the excellence program Statistical methods and models for complex data in San Servolo Island, Venice, Italy.

Professional Memberships

2021–2023 Institute of Mathematical Statistics (IMS).

2018 COST (European Cooperation in Science and Technology) Association.

Fellowships and Awards

2018–2021 Ph.D. scholarship in Statistical Sciences funded by University of Padova, Italy.

2016–2017 Scholarship of the National Program for the Development of Mathematics 2010–2020 of Vietnam Institute for Advanced Study in Mathematics (VIASM).

Award for excellent students, Deparment of Mathematics and Computer Science, Vietnam National University–Ho Chi Minh University of Science.

Research Experiences

Department of Statistical Sciences Paolo Fortunati, University of Bologna, Italy

Aug.-Oct. 2021 ***a short-term project for three months***.

- Objectives of the assignment: (1) develop of efficient algorithms to carry out greedy searches in the space of graphical models with symmetries; (2) implementations of the algorithms through functions to be developed in the statistical environment `R`; (3) application to a set of real data.

Department of Statistical Sciences, University of Padova, Italy

2018–2021 ***Ph.D. student in Statistics***.

- First year: completed the foundation courses on Functional Analysis, Probability Theory, Programming Metodologies, Theory and Methods of Inference, Statistical Models and specialised short courses and seminars.
- Second and third years: research and thesis preparation; Statistical Consulting course.

Ph.D. research: Model selection for colored graphical models for paired data.

- Advisor: Professor Alberto Roverato at the Department of Statistical Sciences, University of Padova.
- Objectives: develop a computationally efficient model search algorithm for the space of colored graphical models for paired data:
  - Review of the literature concerning model selection criteria for the tuning parameter of the graphical lasso (`glasso`) algorithm, and analyse the relationship existing between the degree of sparsity of the graph and value taken by the tuning parameter.
  - Get the ideas of graph coloring, partial orderings, lattices, RCON models and review the structure properties of the lattice structure of models spaces such as the completeness, distributivity, etc.
  - Introduce a novel partial ordering of lattice structure for the class of colored graphical models for paired data which would be fit to the data and easily implemented for operations in model selection exploit the structure properties of model space of graphical models for paired data such as the distributivity and completeness of lattice structure.
  - Construct an efficient model selection procedure, which exploits the structural property of the lattice equiped by the novel ordering into the model searching and respects the principle of coherence, using greedy search method to avoid an exhaustive search.
  - Perform the numerical experiments from the model search procedure applied the novel partial ordering compared to the approach for the submodel relation on lattice structure for the simulated datasets, which contains two sets of homologous variables with multiple different scenarios of the number of vertices, the sparsity and symmetry.
  - Implement of the proposed algorithms through functions to be developed in the statistical environment `R` for the real application to investigate the dynamic activity of each brain region between two hemispheres through changes in the blood oxygenation level dependent (BOLD) signal via functional MRI data.

Institut Denis Poisson, Université d'Orléans, France.

2017–2018 ***Master Thesis: Limits for large random graphs***.

- Advisor: Professor Romain Abraham in the Institut Denis Poisson, Université d'Orléans, France.
- Abstract: Limiting object, called graphon, is a continuum graph where nodes are the elements of $[x, y]$ and the edge between two nodes $x, y \in [0, 1]$ is present with a symmetric probability function $W(x, y)$ $[0, 1]^2$ to $[0, 1]$. The objective is to understand this limiting object, make precise the notion of convergence of growing graphs toward these graphons and exhibit some convergence criteria in the simulation experiments.
- Grade: 17/20.

Vietnam National University–Ho Chi Minh Univeristy of Science, Vietnam.

2016–2017 ***Bachelor Thesis: Multiplicative censoring model***.

- Advisor: Professor Dang Duc Trong in the faculty Mathematics and Computer Science, Vietnam National University-Ho Chi Minh Univeristy of Science, Vietnam.
- Abstract: study the nonparametric estimation using orthogonal projection based on the Laguerre basis of both the density f and the corresponding survival function of $X_i's$ on the multiplicative censoring model $Y_i = X_i U_i, 1 \leq i \leq n$ where $U_i's$ are multiplicative noise with uniform density on $[1 - a, 1 + a]$ for some $a \in (0, 1)$. Only $(Y_i)_{1 \leq i \leq n}$ are observed, and $(U_i)_{1 \leq i \leq n}$ and $(X_i)_{1 \leq i \leq n}$ are independent sequences. Then simulation experiments are implemented by R with different scenarios of distributions of X and tunning parameters in estimation for the dimention of the Laguerre basis.
- Grade: 10/10.

Selected Academic Experiences

Oct. 5th-27th 2021 **Attended a short course "Statistical aspects of Deep Neural Networks" (28 hours)**, *Programme in Economics, Statistics and Data Science, University of Milano-Bicocca, Italy*.

Invited lecturer: Professor Omiros Papaspiliopoulos, Bocconi University, Milano, Italy.

Objectives:

- Covers the foundations of deep neural networks with a special emphasis and priority on more statistical aspects of this research agenda, e.g. high-dimensional regression, the random features models, from PCA to autoencoders, optimisation for neural networks, from deep neural networks to stochastic processes, regularisation, stability and adversarial training, connection between simulation-based inference and neural networks, etc.
- Give an overview of software and implementations in Pytorch.

Sep. 2020 **Joined a course "Regression Modelling with Large Data Sets" (10 hours)**, *Padova, Italy*.

Invited lecturer: Professor Ioannis Kosmidis, Department of Statistics at the University of Warwick and Turing Fellow of The Alan Turing Institute, UK.

Objectives:

- Introduce core theory, methods and software tools for tackling regression problems that involve large data sets in terms of number of observations or in terms of explanatory variables.
- Identify least squares as a core optimisation problem for fitting linear and generalised linear models, and classify the various methods for its solution or regularised versions of it in terms of complexity, memory usage and accuracy.

Apr.-Jun. 2020 **Participated in Statistical Consulting course about the group's title *"Predicting the economic effects of COVID-19 and of the lockdown measures"***.

Instructors: Antonio Canale, Bruno Scarpa, Department of Statistical Sciences, University of Padova, Italy.

Objectives:

- Investigate the effects and the consequences of the deadly virus COVID-19 outbreak and of the political decisions imposed by governments on the level of economic activities and financial market indices through a scientific approach.
- Apply statistical models on collected time series data to obtain reliable measures of the impact of the outbreak on economy and finance.
- Perform sensitive analysis to consider the possible consequences on different scenarios, such as the scenario of a second wave of Corona infections, with the purpose of helping policymaker to make the right decisions on society and economy in future situations.

Languages

Vietnamese	Mother tongue	
English	Fluent	*Excellent in reading, listening, good at writing and speaking.*
Italian	Basic	*Basic words and phrases.*

Computer Skills

Programming Languages	**Advanced R, Python, MATLAB**
Operating Systems	**Ubuntu, macOS**

Referees

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